BEST BRAIN EXAMINATIONS KONSORTIUM
SPECIAL PRIVATE MOCK FOR BECE CANDIDATES – SEPTEMBER 2026
MARKING SCHEME – *SCIENCE*

PAPER TWO

SECTION A [40 Marks]

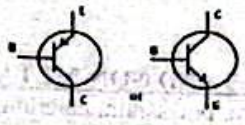
QUESTION ONE

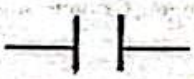
(a) (i) **IDENTIFICATION OF ELECTRONIC COMPONENTS**

A- Transistor; B- Capacitor; C- Resistor; D- Diode

[2 marks @ ½ mark each]

(ii) **DRAWING THE CIRCUIT SYMBOL OF EACH OF THE COMPONENTS**

A- Transistor 

B- Capacitor 

C- Resistor 

D- Diode 

[4 marks @ 1 mark each]

(iii) **FUNCTION OF EACH OF THE COMPONENTS**

A- Transistor

- Acts as a switch to turn circuits on or off.
- Acts as an amplifier - Increases the strength of weak signals

B- Capacitor

- Stores and releases energy
- Smooths out fluctuations in signals.
- Used in timing circuits due to charge and discharge properties

C- Resistor

- Ensures current remains within safe levels.
- Divides voltage within a circuit.
- Converts excess electrical energy into heat

D- Diode

- Allows current in one direction
- Converts AC to DC.
- Prevents reverse current flow.
- Extracts audio signals in receivers

[4 marks @ 1 mark each]

SUBTOTAL = 10 MARKS

(b) (i) **THE PARTS LABELLED I, II, III, IV, V, VI, VII AND VIII.**

I- Mouth; II- oesophagus; III- liver; IV- stomach; V- pancreas;
VI- Small intestines VII- Large intestines VIII- Anus

[8 marks @ 1 mark each]

(ii) **DESCRIBING THE FUNCTIONS OF THE MOUTH IN THE DIGESTION PROCESS**

- Chewing breaks down food into smaller pieces.
- Enzymes in saliva begin the breakdown of starch
- Saliva moistens food, making it easier to swallow.
- The tongue and muscles of the mouth help move food towards the oesophagus

[2 marks @ 1 mark each]

SUBTOTAL = 10 MARKS

CamScanner

(c) (i) **NAME THE MISSING ELEMENTS LABELLED I, II, III, IV, V AND VI.**

I. Beryllium (Be) II - Carbon (C); III - Neon (Ne)
IV- Phosphorus (P) V- Argon (Ar) VI- Potassium (K)

[3 marks @ ½ mark each]

(ii) **LIST TWO ELEMENTS EACH OF *GROUP 2* AND *PERIOD 3.***

Elements of Group 2- Beryllium, magnesium, calcium
Elements of Period 3- Sodium, magnesium, aluminum, silicon, phosphorous, sulfur, chlorine, argon

[2 marks @ ½ mark each]

(iii) **IDENTIFY TWO NOBLE GASES**

Noble gases- Helium, neon, argon

[2 marks @ 1 mark each]

(iv) **CLASSIFYING ELEMENTS UNDER METALS AND NON-METALS**

- *Metals*: lithium, beryllium, sodium, magnesium, potassium, calcium, aluminum
- *Non-metals*: hydrogen, carbon, phosphorous, nitrogen, oxygen, sulfur, fluorine, chlorine, helium, neon, argon.

[3 marks @ 1½ mark each]

SUBTOTAL = 10 MARKS

(d) (i) **IDENTIFICATION OF ANIMALS**

I- Duck; II- Cow; III- Rabbit; IV- Sheep; V- Goat
VI- Fowl; VII- Pig; VIII- Turkey

[4 marks @ ½ mark each]

(ii) **CLASSIFYING EACH ANIMAL AS RUMINANT, POULTRY OR MONOGASTRIC.**

RUMINANT	POULTRY	MONOGASTRIC
Cow, Sheep, Goat	Duck, Turkey, Fowl	Rabbit, Pig

[4 marks @ ½ mark each]

(iii) **DISTINGUISH BETWEEN MONOGASTRIC ANIMALS AND RUMINANTS**

Monogastric animals are ***herbivorous animals*** that have a ***single-chambered stomach*** but are still able to ***digest plant*** material.
Ruminants are four-chambered stomach animals [2 marks @ 1 mark each]

SUBTOTAL = 10 MARKS
TOTAL FOR PART I = 40 MARKS

QUESTION TWO

(a) **DESCRIBING THE WAYS IN WHICH THE STOMACH HELPS IN DIGESTION**

- The stomach muscles contract to churn and mix food.
- Gastric juice enzymes like pepsin break down proteins.
- Hydrochloric acid kills bacteria
- Mucus protects the stomach lining from acids
- It stores food and releases slowly into the small intestine

[4 marks @ 2 marks each]

(b) (i) **USES OF POULTRY.**

- Food- Meat, Egg,
- Manure
- Egg shells
- Feathers
- Waste control
- Entertainment
- Companionship etc

[3 marks @ 1 mark each]

(ii) **CHARACTERISTICS OF RUMINANTS**

- They have four-chambered stomachs
- They regurgitate and rechew their foods

[3 marks @ 1½ mark each]

CamScanner

(c) (i) **NAMING AND EXPLAINING EACH OF THE LABELLED STAGES**

I – Condensation: Water vapor cools and condenses into liquid water droplets, forming clouds in the atmosphere

II –Precipitation: Water vapor in the atmosphere condenses into clouds and falls back to Earth's surface as rain, snow, sleet, or hail

III – Collection: Water gathers in rivers, lakes, and oceans after precipitation, and also accumulates in smaller bodies of water like puddles

IV – Evaporation: Water from oceans, lakes, and rivers is heated by the sun and turns into water vapor, rising into the atmosphere

[4 marks @ 1 mark each]

(ii) **ACTIVITIES THAT CAN AFFECT THE WATER CYCLE**

- Deforestation
- Construction works
- Logging
- Urban development in forested area
- Drawing large amounts of water from the soil,
- Irrigation,
- Soil degradation

etc.

[2 marks @ 1 mark each]

(d) **EXAMPLES OF CONDUCTION**

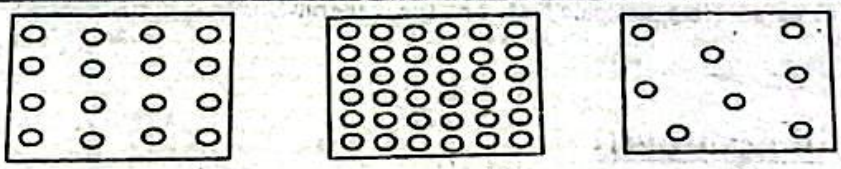
- Hot metal placed in cold water - The metal transfers heat to the water, cooling it down.
- Hot air blown over an ice cube - The warm air transfers its energy to the ice cube, melting it.
- Sitting on a hot surface - Your body transfers heat to the surface you are sitting on, heating it up.
- A metal spoon placed in boiling water - The spoon transfers heat from the water to itself, heating it up quickly.
- Iron and copper pipes connected to a hot water tank - The pipes transfer heat from the tank to the water flowing through the pipes, heating it up.

[4 marks @ 2 marks each]

SUBTOTAL = 20 MARKS

QUESTION THREE

(a) **DRAWING TO SHOW HOW PARTICLES ARE ARRANGED IN THE THREE STATES OF MATTER**



Liquid Solid Gas

[6 marks @ 2 marks each]

(b) (i) **NEWTON'S FIRST LAW OF MOTION**

An object at ***rest will remain at rest***, and an object ***in motion will continue in motion*** at a ***constant velocity*** unless acted upon by ***an unbalanced force***.

[2 marks]

(ii) **CALCULATING MASS OF THE METALLIC BALL.**

Using $F = ma$; $F = 60N$, $a = 3\ m/s^2$, $m = ?$

Mass $= \frac{60}{3} = 20$; therefore, the mass of the metallic ball is **20kg**

[2 marks]

(c) (i) **SAFETY PRECAUTIONS TO CONSIDER WHILE USING THE HOE**

- Wear appropriate clothing, including sturdy shoes and protective gloves, to prevent injuries.
- Keep a safe distance from others while using the hoe to avoid accidental injuries to others.
- Use the hoe with a firm grip and avoid swinging it too close to your body to prevent accidents.
- Watch out for obstacles or hard surfaces that could cause the hoe to bounce back or become difficult control.

[2 marks @ 1 mark each]

CamScanner

(ii) **PROPER WAYS OF HANDLING AGRICULTURAL TOOLS.**

- Clean tools thoroughly after each use to prevent rust and preserve their sharpness.
- Store tools in a dry, covered place (e.g., shed or toolbox) to protect them from the weather.
- Sharpen cutting edges, tighten loose bolts, and lubricate moving parts to ensure efficient use.
- Use each tool for its intended purpose to prevent damage to the tool and injury to the user.
- Avoid using worn-out tools in poor condition to prevent accidents.
- Agrochemicals should be used by wearing gloves and masks to prevent poisoning.
- Wear hand gloves, boots and mask while sprinkling pesticides

[4 marks @ 1 mark each]

(d) (i) **IDENTIFYING THE BREEDING PLACES OF HOUSEFLIES**

Rotten organic matter, Manure, still water, Trash bins, Sewers and drains, Rotting plants.

[2 marks @ ½ mark each]

(ii) **NAMING THE DISEASES TRANSMITTED BY THE HOUSEFLY**

Cholera, Typhoid, Dysentery, Salmonellosis, E coli.

[2 marks @ 1 mark each]

SUB-TOTAL = 20 MARKS

QUESTION FOUR

(a) (i) **CHARACTERISTICS OF ANIMALS FOUND IN DESERTS**

- **Water conservation** – Desert animals conserve water by producing highly concentrated urine and dry faeces (e.g. camels, kangaroo rats).
- **Nocturnal behaviour** - Many are active at night to avoid extreme daytime heat (e.g. desert foxes, snakes).
- **Burrowing habits** - Some animals live in burrows to escape the heat and maintain moisture (e.g. desert tortoises, jerboas).
- **Light-coloured bodies** – Reflect sunlight and reduce heat absorption (e.g. fennec fox, pale lizards).
- **Ability to withstand heat** – Some can tolerate high body temperatures without sweating or panting (e.g. camels).
- **Efficient metabolism** – They can generate water through the digestion of food (metabolic water).
- **Fat storage** – Some store fat in specific body parts to convert to water and energy when needed (e.g. camel's hump).
- **Long limbs or ears** – Help dissipate body heat (e.g. jackrabbits, fennec foxes).

[2 marks @ 1 mark each]

(ii) **IMPORTANCE OF LANDFORMS**

- **Habitat:** Landforms provide habitats for various plants and animals.
- **Water Resources:** Landforms like mountains and valleys play a crucial role in the water cycle.
- **Agriculture:** Different landforms have different soil types and fertility levels, which can affect agricultural productivity.
- **Recreation and Tourism:** Landforms such as mountains, canyons, and islands attract tourists and provide opportunities for recreational activities like hiking, skiing, and water sports.
- **Geological Resources:** Landforms can contain valuable geological resources such as minerals, fossil fuels, and precious metals.
- **Natural Protection:** Landforms like hills and mountains can act as natural barriers, protecting areas from extreme weather events like hurricanes.

[2 marks @ 1 mark each]

(b) **USES OF METALS**

- Gold is used for making ornament and jewelry
- Mercury is used as a liquid in a thermometer
- Magnesium is combined with Aluminium to make aircraft bodies
- Iron is used for constructing building, bridges, car bodies and engines
- Silver is used for making cutlery **[3 marks @ 1 mark each]**

CamScanner

USES OF NON-METALS

- Nitrogen is used in manufacturing ammonia.
- Helium is used to fill balloon.
- Phosphorus is used for fertilizer.
- Chlorine used in treatment of water.
- Oxygen is used for combustion(burning).
- Argon in fluorescent lamp.
- Carbon is used as a clue to determine the age of an object.

[3 marks @ 1 mark each]

(c) **DESCRIPTION OF ENERGY CHANGES**

(i) Chemical energy → kinetic energy → sound + heat energy

(ii) Potential energy → kinetic energy → Mechanical energy → electrical energy

(iii) Mechanical energy → sound energy → heat energy

[6 marks @ 2 marks each]

(d) (i) **DIFFERENCE BETWEEN BIODEGRADABLE AND NON-BIODEGRADABLE WASTES**

Biodegradable Wastes: Materials that can be broken down naturally by microorganisms (bacteria, fungi) into simpler substances over time. They decompose and return to the environment.

***Examples*:** Food scraps, garden waste, paper, cotton.

Non-biodegradable Wastes: Materials that cannot be broken down by natural processes or take an extremely long time to decompose. They accumulate in the environment.

***Examples*:** Plastics, glass, metals, synthetic fibers.

[2 marks @ *1 each for explanation and 1* for *a correct example*]

(ii) **DIFFERENCE BETWEEN HAZARDOUS AND NON-HAZARDOUS WASTES.**

Hazardous Wastes: Wastes that pose a substantial danger to public health or the environment. They require special handling and disposal. **Examples:** Used batteries, old paints, pesticides, medical waste, corrosive cleaning agents.

Non-hazardous Wastes: Wastes that do not pose an immediate threat to human health or the environment. They are typically safe for general disposal methods.

Examples: Food wrappers, paper, cardboard, plastic bottles (non-toxic).

[2 marks @ *1 each for explanation and 1* for *a correct example*]

SUBTOTAL = 20 MARKS

QUESTION FIVE

(a) **DIFFERENCE BETWEEN THE FOLLOWING FARMING SYSTEMS;**

(i) **LAND ROTATION AND SHIFTING CULTIVATION;**

Land Rotation: Farmers cultivate a piece of land for a few years, then leave it fallow to recover fertility while moving to another part of their *own* land. They eventually return to the original plot.

Shifting Cultivation: A farming method where farmers cultivate a piece of land for a few years until its fertility declines, then abandon it to clear and cultivate a *new* piece of land, often relocating their settlement.

[2 marks @ 1 mark each]

(ii) **MONOCULTURE AND MONO CROPPING;**

Monoculture: Refers to the practice of growing a single crop species in a large area over several seasons or years.

Monocropping is growing the same crop on the same land *season after season* without rotation.

[2 marks @ 1 mark each]

(iii) **MIXED CROPPING AND MIXED FARMING.**

Mixed Cropping: The practice of growing two or more different crops simultaneously on the same piece of land in the same season

Mixed Farming is an agricultural system that combines both crop cultivation and livestock rearing (e.g., cattle, poultry) on the same farm. **[2 marks @ 1 mark each]**

(b) (i) **REASONS WHY ECOSYSTEM MUST BE PRESERVED**

- It regulates the essential ecological processes
- Supports life systems and stability.
- Responsible for the cycling of nutrients between living and non-living.

CamScanner

- Maintains a balance among the various trophic levels in the ecosystem.
- It cycles the minerals through the biosphere.
- The abiotic components help in the synthesis of organic components that involves the exchange of energy

[3 marks @ 1 mark each]

(ii) **ACTIVITIES THAT DISTURB THE BALANCE IN AN ECOSYSTEM**

- burning of wood
- excessive release of CO_2 through cars
- industrial release of excess CO_2 into air
- deforestation

[3 marks @ 1 mark each]

(c) **HOW A SEAT BELT OPERATES**

- The seat belt provides the unbalanced force that prevents the driver from its continuous uniform straight-line motion, hence keeping him safe from the sudden stop of the car.

[3 marks]

(d) (i) **USES OF SOLIDS IN EVERYDAY LIFE**

- Building houses
- Constructing of roads
- Making jewellery
- Making kitchenware
- Manufacturing of electronic equipment
- Building of machinery
- Etc.

[2 marks @ 1 mark each]

(ii) **EXAMPLES OF LIQUIDS THAT ARE USEFUL IN OUR DAILY LIFE**

- Water – essential for our survival as humans as it is involved in many of the body's processes
- Milk – a typical source of protein, calcium and vitamins
- Oil – for cooking, fuelling vehicles and as raw materials for the production of many products
- Juice – refreshment and popular choice of minerals and vitamins
- Alcohol – a recreational beverage which is also used as solvents in many products
- Gasoline – important source of fuel for vehicles
- Coffee – a popular beverage
- Cleaning solutions (vinegar, soap, bleach etc.) – used for cleaning

[3 marks @ 1 mark each]

SUBTOTAL = 20 MARKS

TOTAL FOR PART II = 60 MARKS

THUS TOTAL FOR PAPER 2 = (PART I + PART II) = 100 MARKS

PAPER I [40 MARKS]

1. B	11. B	21. C	31. A
2. B	12. C	22. C	32. B
3. D	13. A	23. A	33. A
4. C	14. D	24. C	34. B
5. B	15. C	25. B	35. C
6. C	16. C	26. B	36. A
7. C	17. A	27. C	37. A
8. A	18. A	28. D	38. A
9. D	19. C	29. D	39. A
10.A	20. B	30. D	40. C

GRAND TOTAL = PAPER 1 (40) + PAPER 2 (100) = 140 MARKS

$$\text{OVERALL SCORE} = \frac{\text{TOTAL SCORE}}{140} \times 100$$

CamScanner